3719. Longest Balanced Subarray I

Solved

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You are given an integer array `nums`.

A **subarray** is called **balanced** if the number of **distinct even** numbers in the subarray is equal to the number of **distinct odd** numbers.

Return the length of the **longest** balanced subarray.

Example 1:

> **Input:** `nums = [2,5,4,3]`
>
> **Output:** 4
>
> **Explanation:**
>
> - The longest balanced subarray is `[2, 5, 4, 3]`.
> - It has 2 distinct even numbers `[2, 4]` and 2 distinct odd numbers `[5, 3]`. Thus, the answer is 4.

Example 2:

> **Input:** nums = [3,2,2,5,4]
>
> **Output:** 5
>
> **Explanation:**
>
> - The longest balanced subarray is `[3, 2, 2, 5, 4]`.
> - It has 2 distinct even numbers `[2, 4]` and 2 distinct odd numbers `[3, 5]`. Thus, the answer is 5.

Example 3:

> **Input:** nums = [1,2,3,2]
>
> **Output:** 3
>
> **Explanation:**
>
> - The longest balanced subarray is `[2, 3, 2]`.
> - It has 1 distinct even number `[2]` and 1 distinct odd number `[3]`. Thus, the answer is 3.

Constraints:

- `1 <= nums.length <= 1500`
- `1 <= nums[i] <= 10^5`

Python:

```
class Solution:
    def longestBalanced(self, nums: List[int]) -> int:
        n = len(nums)
        res = 0

        seen = [0] * (max(nums) + 1)

        for i in range(n):
            A = [0, 0]
```

```
            for j in range(i, n):
                val = nums[j]
                if seen[val] != i + 1:
                    seen[val] = i + 1
                    A[val & 1] += 1

                if A[0] == A[1]:
                    res = max(res, j - i + 1)

        return res
```

JavaScript:

```
const longestBalanced = nums => {
    const n = nums.length;
    let res = 0;

    const seen = new Int32Array(Math.max(...nums) + 1);

    for (let i = 0; i < n; i++) {
        const A = [0, 0];

        for (let j = i; j < n; j++) {
            const val = nums[j];
            if (seen[val] !== i + 1) {
                seen[val] = i + 1;
                A[val & 1]++;
            }

            if (A[0] === A[1])
                res = Math.max(res, j - i + 1);
        }
    }

    return res;
};
```

Java:

```
class Solution {
    private static int[] seen = new int[100001];
    private static int leet = 0;
    public int longestBalanced(int[] nums) {
        leet++;
        int n = nums.length;
        int res = 0;
```

```
        for (int i = 0; i < n; i++) {
            int[] A = new int[2];
            int marker = (leet << 16) | (i + 1);
            for (int j = i; j < n; j++) {
                int val = nums[j];
                if (seen[val] != marker) {
                    seen[val] = marker;
                    A[val & 1]++;
                }

                if (A[0] == A[1])
                    res = Math.max(res, j - i + 1);
            }
        }

        return res;
    }
}
```